BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Paper 1: Data Representation

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What does the MDR hold?

2. grid = [[11,12],[21,22]]. With indexes starting at 0, what is grid[1,0]?
   A) 21
   B) 11
   C) 12
   D) 22

3. "Changing a variable name automatically fixes every logic error." Correct this. Explain why.

4. Two functions each have a local total. Several procedures share a global score. Correct both claims. Explain why. (a) All subprograms can read every local variable. (b) Only one subprogram can read a global variable.

Today's objectives

Explain the core ideas of paper 1: data representation using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Binary place values

Use 128,64,32,16,8,4,2,1 for unsigned eight-bit numbers.

Hexadecimal

Group binary digits into nibbles of four bits.

Image size

Width x height x colour depth yields uncompressed bits.

Sound size

Samples/second x seconds x bits/sample yields mono uncompressed bits.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Binary place values
Use 128,64,32,16,8,4,2,1 for unsigned eight-bit numbers.

Hexadecimal
Group binary digits into nibbles of four bits.

Image size
Width x height x colour depth yields uncompressed bits.

Sound size
Samples/second x seconds x bits/sample yields mono uncompressed bits.

I Do - Worked model

Convert eight-bit 10110110 to denary and hexadecimal.

- Add values: 128+32+16+4+2=182.
- Split 1011 and 0110.
- Convert to B6 hexadecimal.
- Check 11 x 16+6=182.

We Do - Guided practice

1. Calculate the bits in a 60 x 40 bitmap with four-bit colour depth.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Correct the raw data size conversion from bits to bytes.

```
width = 20
height = 10
depth = 8
bytes = width * height * depth
print(bytes)
```

You Do - Independent exam questions

Q2. Convert hex 2A to denary. [3 marks]

Q3. Convert eight-bit binary 11001100 to hex. [2 marks]

Q4. Calculate mono sound bytes at 1000 samples/s, 4 s, 8 bits/sample. [4 marks]

Exit Ticket - True or False?

One hexadecimal digit can represent four binary bits.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Data or an instruction being transferred to or from memory.

2. A - 21

3. Clear names show what values mean and help people read and change code.

4. (a) Each local total is used only in its own scope. (b) A global score can be used across the program. This risks unwanted changes. Use arguments and return values to make changes clear.

I Do - Explain the model

Add values: 128+32+16+4+2=182.

Split 1011 and 0110.

Convert to B6 hexadecimal.

Check 11 x 16+6=182.

We Do - Model answers

1. Divide eight to get 1200 bytes if requested.

Code First - Answer

```
width = 20
height = 10
depth = 8
bytes = width * height * depth DIV 8
print(bytes)
```

Expected output: 200

You Do - Self-mark

Q2: 2 x 16+10=42.

Q3: CC.

Q4: 1000 x 4 x 8 / 8=4000 bytes.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.